VN Equities - Barra-style Risk Model

Cross-sectional factor risk model | VN100 universe | 2021-2026 | 242 weekly fits

1. Overview

A Barra-type cross-sectional risk model for the Vietnamese stock market, built alongside the existing Fama-French VN-3/VN-4 academic factors. Each week we regress stock returns on per-stock exposures (8 style factors + industry dummies) to estimate factor returns and a factor covariance matrix, then decompose portfolio risk. Goal: risk measurement/forecasting, not an alpha claim. Data is real (vnstock); no look-ahead (price descriptors use data up to t; fundamentals lag 45 days).

2. Methodology

Model per week: r_i = f_country + SUM_m X_ind(i,m)*f_ind,m + SUM_s X_style(i,s)*f_style,s + u_i,
with identification constraint SUM_m capwt_m * f_ind,m = 0 and regression weights sqrt(market cap), solved exactly via the KKT system. Styles standardised cross-sectionally each week (cap-weighted mean 0, std 1, winsorised +/-3 MAD): SIZE, BETA, MOMENTUM (12-1m), RESVOL (120d), LIQUIDITY (60d turnover), VALUE (EP+BM), LEVERAGE (Debt/Equity), QUALITY (ROE). Industries (>=4 names): Banks, Real estate, Securities, Utilities, Food & beverage, Construction, Building materials, Plastics-chemicals, Transport, Other.

3. Factor returns (annualised) and significance

Style	Ann.ret %	Ann.vol %	t-stat	IR
SIZE	-0.23	9.44	-0.05	-0.02
BETA	-5.20	14.86	-0.75	-0.35
MOMENTUM	-1.72	9.87	-0.37	-0.17
RESVOL	+5.04	13.26	+0.82	+0.38
LIQUIDITY	-4.81	13.40	-0.77	-0.36
VALUE	-1.05	9.82	-0.23	-0.11
LEVERAGE	-1.51	7.33	-0.45	-0.21
QUALITY	-3.92	6.59	-1.28	-0.60

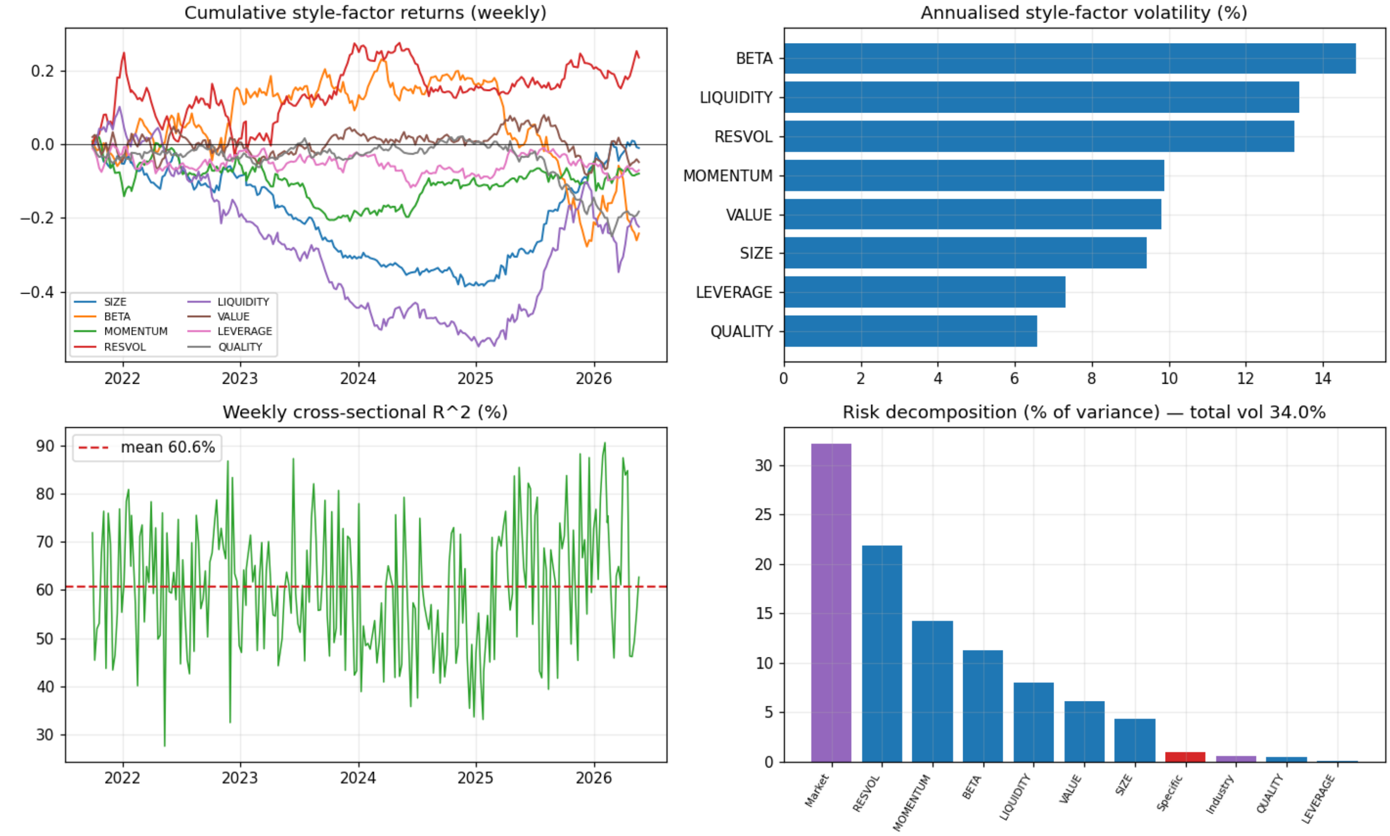
Average cross-sectional R^2 = 60.6% (factors explain a large share of return dispersion). No style premium is statistically significant over this window (all |t| < 1.3) - expected for a thin VN100 universe over ~5 years. This is a risk model, not an alpha claim.

4. Portfolio risk decomposition (equal-weight, latest week)

Total annualised volatility = 34.0% (factor 33.8% / specific 3.3%). Idiosyncratic risk is almost fully diversified; risk is dominated by Market + common style factors.

Source	% of variance
Market	+32.2%
RESVOL	+21.9%
MOMENTUM	+14.3%
BETA	+11.3%
LIQUIDITY	+8.0%
VALUE	+6.1%
SIZE	+4.4%
Specific	+1.0%
Industry (all)	+0.5%
QUALITY	+0.4%
LEVERAGE	+0.0%

5. Charts



6. Honest caveats / next steps

- Only ~47 names usable (vnstock community rate-limit capped the fetch at 79/100); effective ~46-47 stocks/week after a 1-year history filter - thin for ~19 factors, so estimates are noisy.
- BETA uses a median-market proxy; fundamentals are lagged quarterly.
- Covariance is the sample estimate; production would add EWMA + Newey-West and bias-statistic backtests (does forecast vol match realised?).
- Next: expand to HOSE (~400 names) for a firmer cross-section; add factor-risk attribution for live portfolios.